Placeholder

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Abstract

Write a concise summary of your project here (usually 150-250 words). It should state the research question, briefly describe the data, mention the primary linear regression models used, and summarize the key findings and conclusions.

Introduction

Data and Exploratory Analysis

Our transit data comes from the Transit Costs Project collaboration at the NYU Marron Institute of Urban Management. The express purpose of this dataset is to document the costs of transit projects across the world for international comparison. The transit projects under consideration all involve a railway, including light rail, trams, rapid transit/metro systems, and heavy rail (the meaning of this terminology changes based on local vernacular). This means busses, ferries, rideshare, bikes or other forms of transportation are not included in the dataset. This data was compiled by academic researchers and documents transit projects from . Each observation is a unique transit project undertaken by that country

To account for spatial factors, country shapefiles from a public ArcGIS dataset were merged into the . The map projection used in the dataset is the Mercator projection, which emphasizes accuracy of shape over size. The final dataset that was merged with the transit dataset was a cross-cultural psychology dataset from social psychologist Geert Hofstede. This dataset includes Hofstede's rankings of countries along six cultural dimensions: individualism-collectivism, uncertainty avoidance, strength of social hierarchy (called 'power-distance'), motivation toward achievement and success (formerly masculine v. feminine), and long-term vs. short term orientation. The observations in the Hofstede dataset were individual countries (or groupings of countries such as "Arab states").

We remove a number of variables such as `source` and `currency` that serve only to tell us the source of the data or the local currency, so we can work with variables relevant to our questions such as `real_cost` and `length`. We then set the `real_cost` variable to be a number, and filter out observations with missing variables or that have yet to be completed (as we do not care for projects yet to begin or yet to finish).

We create three new variables: `duration` (dbl): how long the project took, computed as the difference between the end and start years `region` (character): which geographic region the transit project's country is located in, as per the country's ISO code `group` (character): a further grouping of projects by geographic regions, to ensure that there are sufficient observations to represent different geographic areas

Introduction

Project Goal

To analyze and compare transit projects around the world to understand the factors that affect the cost and speeds of projects around the world, including the additional impact of geographic or cultural factors.

Research Questions

1. Project Speed What variables are significant when predicting the speed at which transit projects are completed?
2. Project Cost Which variables are significant when predicting the cost of a transit project?
3. Spatial and Cultural factors Are spatial and cultural factors significant for models that predict the cost and speed of a project?

Methodology

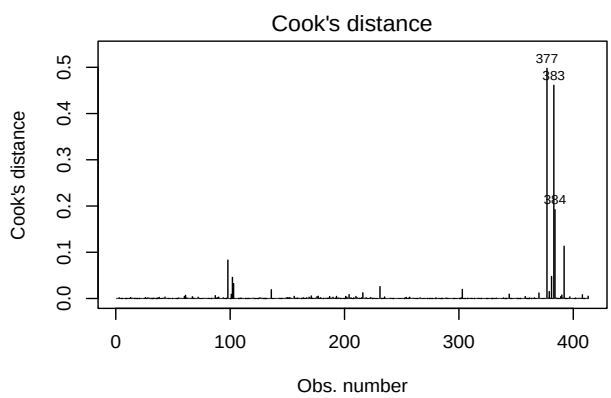
Three models were fit to the combined data. Each of these models tackle each of our connected research questions.

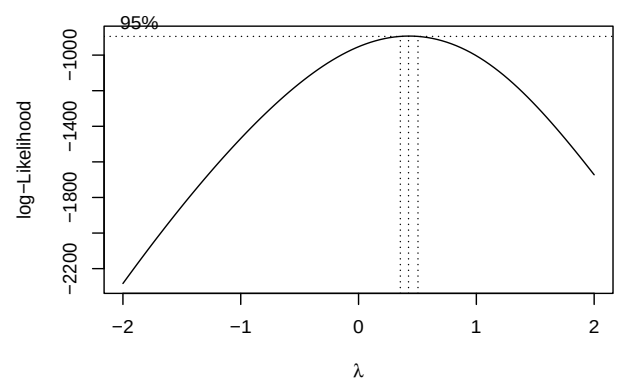
All of our models encode

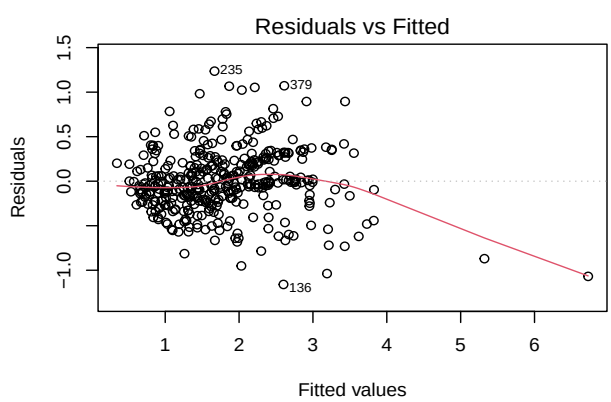
Model 1 - Project Speed

$$Y_i = \beta_0 + \beta_1 \cdot X_{1i} + \beta_2 \cdot X_{2i} + \beta_3 \cdot X_{3i} + \sum_{j=4}^{9} \beta_j \cdot X_{ji}$$

Where:
X_{1i} = the natural log of the real cost of the project in millions of USD X_{2i} = length of the project in km X_{3i} = indicator of whether the project is based on railroad or not (0 if not, 1 if yes) And the sum notes the effect of the regional group the project it belongs to. There are 7 regional groups, with the base case being the Americas



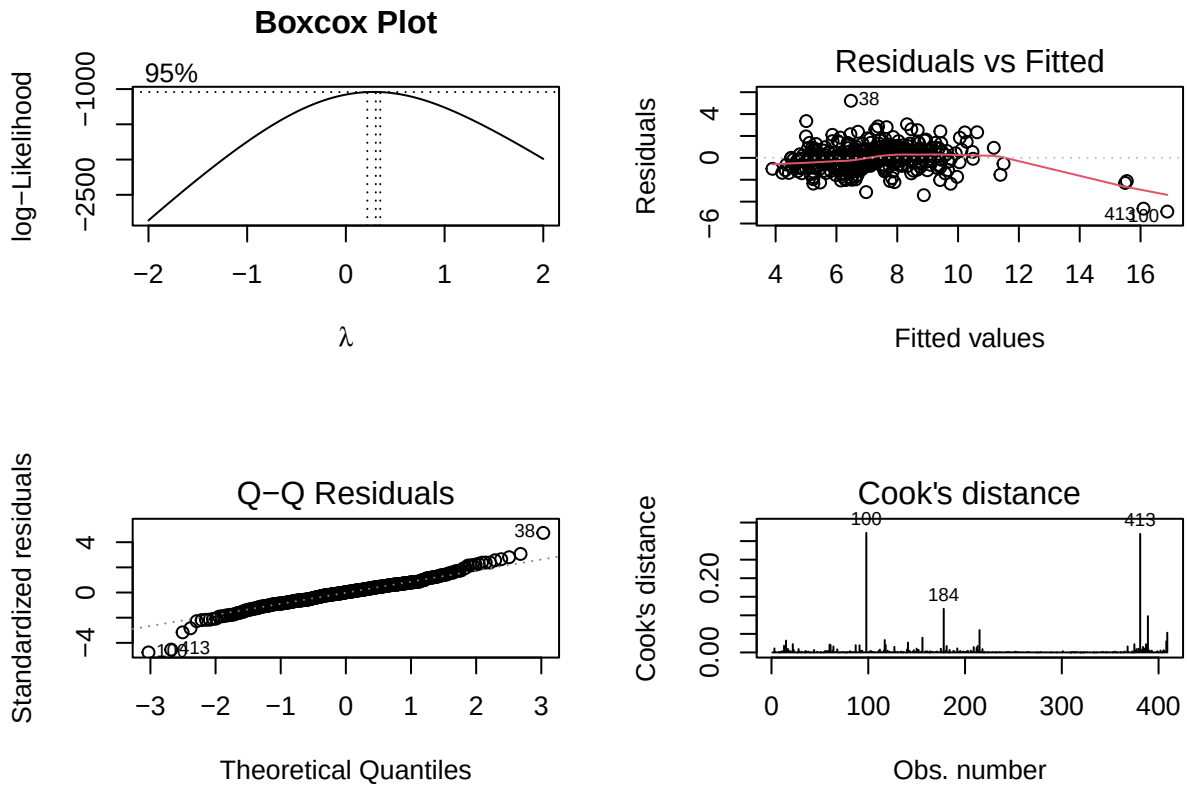




Model 2 - Cost Prediction

$$Y_i^{\frac{1}{4}} = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + \beta_5 X_{5i} + \beta_6 X_{6i} + \beta_7 X_{7i} + \beta_8 X_{8i} + \beta_9 X_{9i} + \beta_{10} X_{10i} + \beta_{11} X_{11i}$$

Where: Y_i = The real cost of a project. Real cost is the nominal cost adjusted based on inflation indexes between countries. $X_{1i}, X_{2i}, \ldots, X_{6i}$ are dummy variables for the region group the country belongs in. X_{7i} = Midpoint year of construction for the project X_{8i} = Length of transit route built (km) X_{9i} = Number of stations built X_{10i} = Purchasing Power Parity (PPP), based on midpoint of construction X_{11i} = Duration of transit project (years)



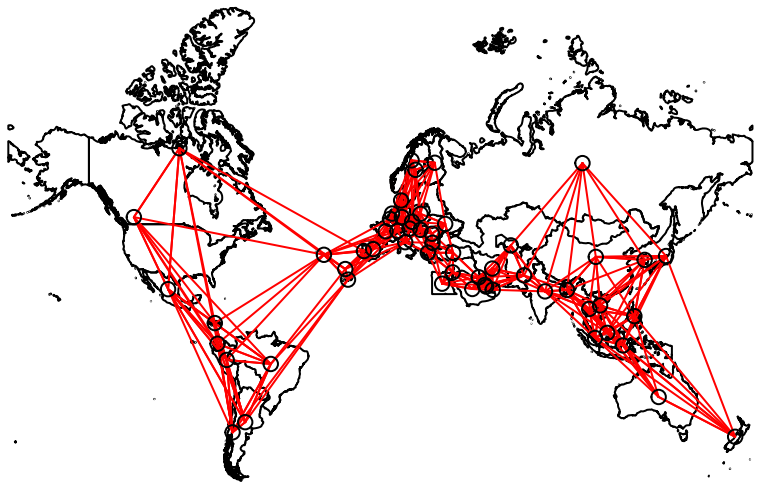
To address non-constant variance and improve residual normality, a Box–Cox transformation was applied to the response variable. This subset of covariates was chosen through backward stepwise regression using AIC as the selection criteria. Cook's distance was used to identify outliers and VIF was used to assess multicollinearity but neither outliers or multicollinearity were identified in this model.

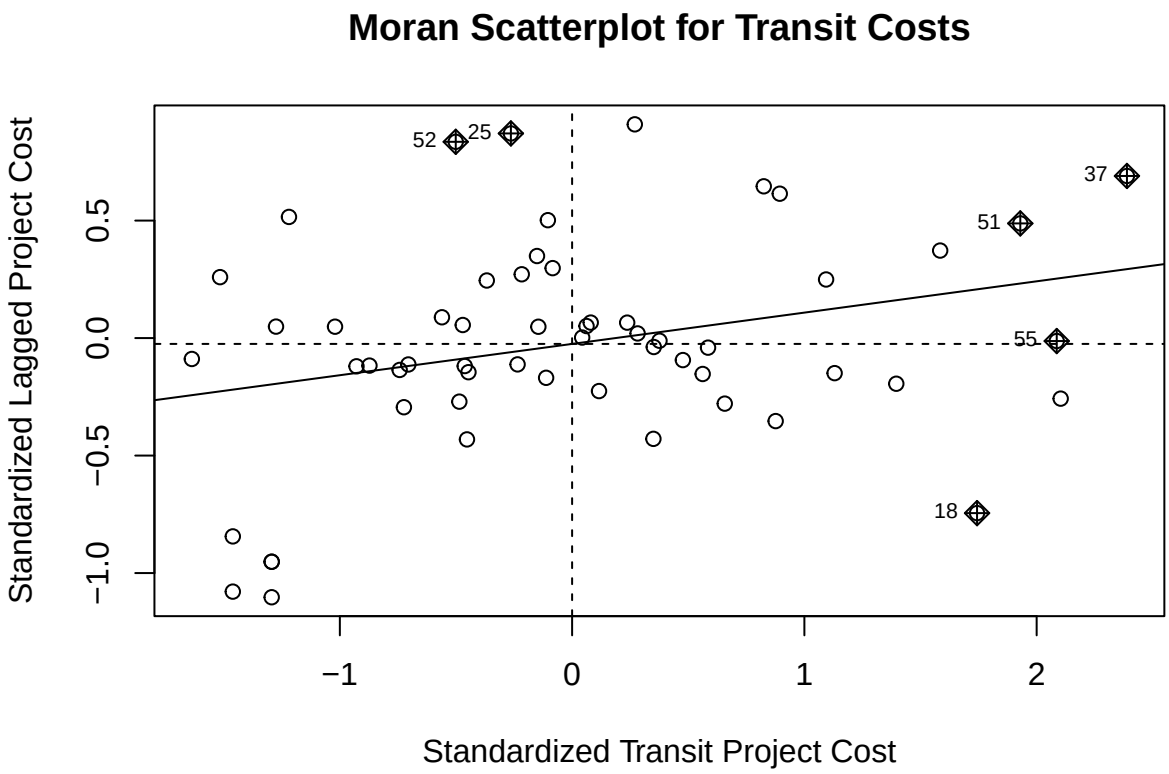
Accounting for spatial

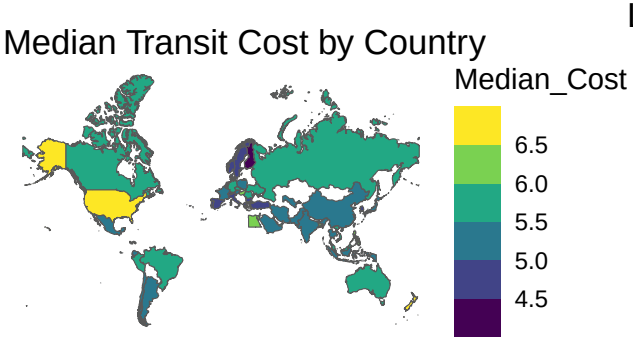
A particular concern for our dataset is spatial autocorrelation. There is strong reason to believe that the presence of transit in neighboring countries could disincentivise or incentivise countries to build it themselves. If a country's passenger rail buildout is successful, it encourages neighbors to invest in rail projects themselves. Certain countries have legal frameworks or cultural biases that discourage infrastructure investments like transit, and these features are often shared among neighboring countries. Spatial autocorrelation may also capture the aspects of geography that make it more costly or inexpensive to build in certain regions, such as soil quality or climate.

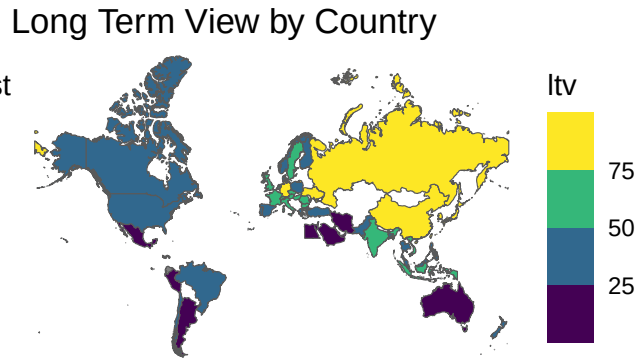
To test for spatial autocorrelation, we would control for

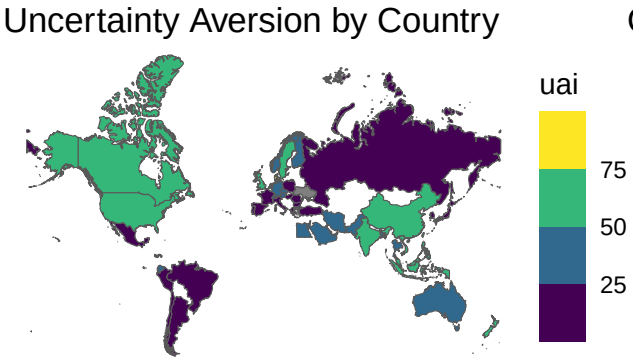
```r
plot(st_geometry(medshp))
plot(dist_nb, coords = st_coordinates(centroid), add = TRUE, col = "red")
```




Moran Scatterplot for Transit Costs
Standardized Lagged Project Cost
Standardized Transit Project Cost
52
25
37
51
55
18
0.5
0.0
−0.5
−1.0
−1
0
1
2







Some econometric procedures recommending accounting for spatial autocorrelation by inflating standard errors.

Model 3 -

Model 4 -

Results

Table 1: Summary of the transformed regression model.

term	estimate	std.error	p.value
(Intercept)	-87.902	20.978	0.000
groupAsia	-0.503	0.239	0.036
groupEastern Europe	-0.897	0.328	0.007
groupEurope	-1.702	0.248	0.000
groupNorth Africa & West Asia	-0.816	0.294	0.006
groupSE Asia & Oceania	0.638	0.310	0.040
groupSouthern Asia	-0.813	0.315	0.010
year	0.047	0.010	0.000
length	0.049	0.006	0.000
stations	0.027	0.008	0.001
ppp_rate	-0.149	0.082	0.068
duration	0.069	0.020	0.001

r.squared	adj.r.squared	sigma
0.694	0.686	1.115

Conclusion

Appendix